① srcML of lines 14–15 and 18–20

```
<for pos:start="14:5">g for a<control><expr><name>qubit</name></expr><range>in <expr><call><name>
    ↪ range</name><argument_list>(<argument><expr><name>nb</name><operator>-</operator>
    ↪ <literal type="number">1</literal></expr></argument>)</argument_list></call></expr></range>
    ↪ </control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>cxc</name></name>
            ↪ <argument_list>(<argument><expr><name>qubitd</name></expr></argument>, <argument><expr>
            ↪ <name>qubit</name><operator>+</operator><literal type="number">1</literal></expr>
            ↪ </argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></for>

<expr_stmt pos:start="18:1">g<expr><call><name><name>qc</name><operator>.</operator><name>measurec
    ↪ </name></name><argument_list>(<argument><expr><name>nd</name></expr></argument>, <argument><expr>
    ↪ <name>ne</name></expr></argument>)</argument_list></call></expr></expr_stmt>
<with>f with <expr><call><name><name>qc</name><operator>.</operator><name>if_testf</name></name>
    ↪ <argument_list>(<argument><expr><tuple>(<expr><name><name>qc</name><operator>.</operator><name>
    ↪ clbitse</name><index>[<expr><name>n</name></expr>]</index></name></expr>, <expr>
    ↪ <literal type="number">1</literal></expr>)</tuple></expr></argument>)</argument_list></call>
    ↪ </expr><block>:<block_content>
    <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>xc</name></name>
        ↪ <argument_list>(<argument><expr><literal type="number">0d</literal></expr></argument>)
        ↪ </argument_list></call></expr></expr_stmt>
</block_content></block></with>
```

a `for` → unrolled **b** range → iteration count, from line 4 **c** gate
d qubit argument, evaluated per iteration **e** classical bit, written by the measurement and read by the condition
f `if_test` → conditioned block **g** position → source line

② Trace

line	iteration	operation
12, 13	every	12 rotations
15	0, 0	cx q_0, q_1
15	0, 1	cx q_1, q_2
15	1, 0	cx q_0, q_1
15	1, 1	cx q_1, q_2
17	–	h q_3
18	–	measure $q_3 \to c_3$
20	–	x q_0 if c_3

③ Slice of q_0

